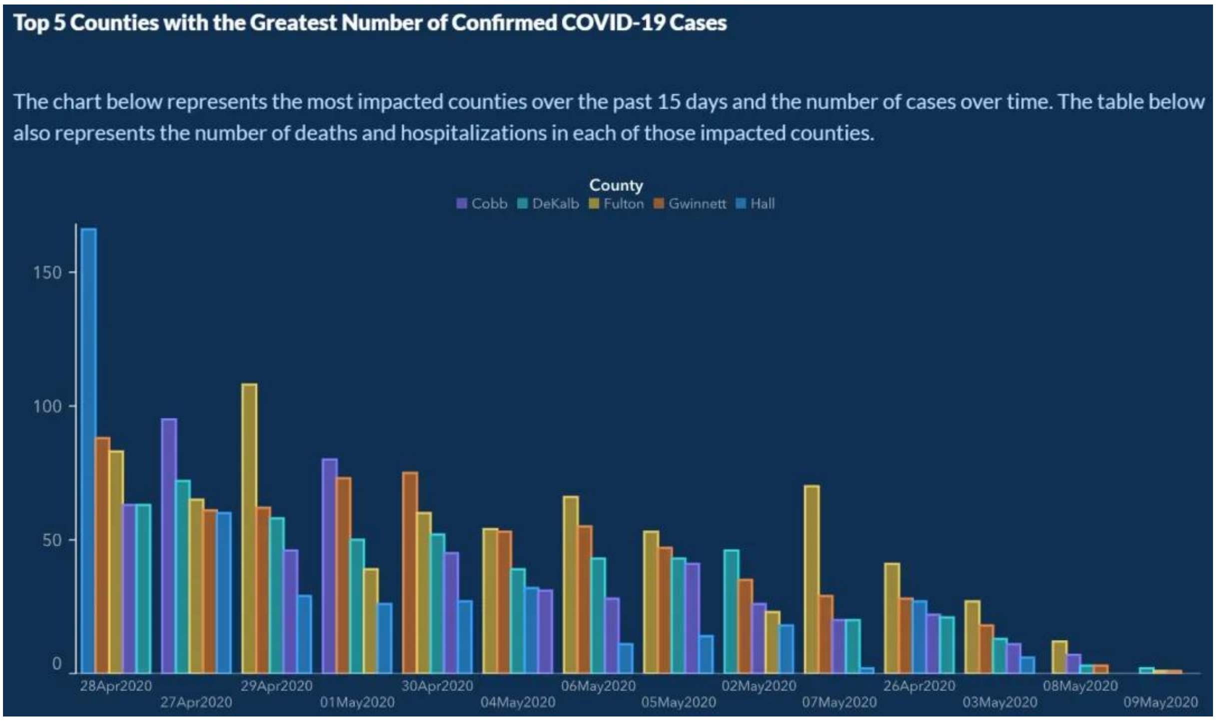
Bad Data Visualization Example: Georgia COVID-19 Cases

Source: Georgia Department of Public Health (GDPH),published in May 2020.*(Reported by The Atlanta Journal-Constitution and Vox)*

Reference URL: https://www.vox.com/covid-19-coronavirus-us-response-trump/2020/5/18/21262265/georgia-covid-19-cases-declining-reopening

Visual:



2. The Problem (Analysis)

This bar chart,published by the Georgia Department of Public Health to illustrate the daily count of new COVID-19 cases in the top five counties, contains a critical error in axis ordering, leading to severe misinterpretation.

- **Non-Chronological X-Axis:** The fundamental problem is that the X-axis (Time) is not ordered chronologically. Instead of following the natural progression of dates (e.g., April 26, April 27, April 28), the dates are shuffled. For instance, "May 2" is followed by "April 26," which is then followed by "May 3."
- **Manipulation of Trends:** The dates appear to have been sorted by the **magnitude of the values** (descending order) rather than by time. This arrangement artificially creates a smooth, downward slope, giving the false impression that the number of cases was steadily declining over the selected period.
- **Statistical Violation:** Time is an ordinal variable that must preserve its sequence to be meaningful. Treating time as a nominal variable (reordering it arbitrarily) destroys the temporal integrity of the data and misleads the public regarding the pandemic's trajectory.

The Solution

To correct this visualization and present the data honestly, the following steps should be taken:

- **Chronological Ordering:** The X-axis must be strictly sorted by date in ascending order (past to present). This is non-negotiable for time-series data.
- **Chart Type Selection:** Instead of a "clustered bar chart," which becomes cluttered and hard to read with multiple categories (counties) over time, a Line Chart should be used.
- **Trend Analysis:** Since daily data can be noisy, adding a 7-day moving average line for each county would help the audience visualize the actual trend (whether it is flattening, rising, or falling) without the distraction of daily fluctuations.